

Brenton's Buccaneers



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EXECUTIVE SUMMARY

The premise behind this design challenge is to create a boat that will be subjected to multiple one on one battle rounds in a freshwater body. It is also important to understand only one boat can come out victorious, so strategy and adaptability is critical to success. A good boat can operate well consistently in a controlled environment. A great boat can thrive regardless of their opponent.

Every group had approximately 15 weeks to complete this mighty feat. The design competition provides the opportunity for students early on in their engineering degree to get hands on experience, apply their education, and give every student the opportunity to create a device they are proud of. As with anything, the best way to become better is to make mistakes and learn from them. Teams that can succeed at rapid prototyping, testing, and adjusting are the ones that are most capable of creating a competitive boat. The key foundation behind this competition is based on the ability to fail fast and learn often. Since most students in this class have never had to create something from design to product, flaws are inevitable. What defines a group's success are factors like group dynamic, time management, and perseverance. The real purpose of this project is to have students practice and learn these principles.

Though the competition has concluded, the impacts it had on every engineer participating will exist for the rest of their personal and professional life. The importance of being able to take on the challenge of brainstorming, evaluating, prototyping, and testing solutions within a design process is something any engineer will have to do in their career, regardless of the field they are going into. Ethical decision making was also a major part of this course and is immensely important in the context of any engineering environment. Since a huge concern was safety in this aquatic environment every team had to choose when prioritize feasibility when executing innovative ideas. The importance of ethical judgement always comes up in biomedical

engineering because often devices are being created that directly impact people's lives. The benefit of taking risks to try something new cannot be held in higher regards than people's safety.

Through this design competition, many lessons were learned in both the technical and non-technical areas of engineering. It is guaranteed that every person actively participating in this challenge learned a lot about themselves as well as creating something from start to finish.

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INTRODUCTION

Barbie is infamous for her many careers and experiences throughout her lifetime. She has always been a trendsetter ahead of her time who can face anything thrown her way, but this time she is dealing with her most difficult obstacle yet. Barblandia is under attack by the cunning Captain Bechara Blackwater and his swashbuckling scallywags: Admiral Abyssal Ani and Abbie the Aquatic Assassin. This cutthroat crew has sent an army of 21 different watercrafts ahead of them, each with unique and unknown capabilities. To beat her nautical nemeses, Barbie decided to recruit the top mechanical engineering students at Colorado State University to create a valiant vessel capable of neutralizing and capturing these ocean outlaws. The fate of Barblandia rests in

Brenton's Buccaneers' ability to meticulously complete both the design analysis and solution stages of this task before Captain Bechara docks on the dawn of the second of December.

Barbie conveyed the importance of this mission clearly. Brenton's Buccaneers must take an iterative design approach to create an unsinkable boat capable of maneuvering across the ocean and defeating any adversaries in their path. If successful, this boat can be immortalized as the saving grace of Barblandia; if unsuccessful, Barbies everywhere could be destroyed.

Brenton's Buccaneers were faced with challenges that included programming this ship with radio frequency, waterproofing every component, safely seating the Captain and First Mate Barbie during their battles, and operating under a looming deadline.

In the end, Brenton's Buccaneers were able to create a beautiful wooden craft that could be controlled all over the ocean and was never sunk by any opponents. Brenton's Buccaneers had chosen to prioritize the boat's mobility and safety over weaponry. Because of all the distances Barbie was able to travel, she reached Captain Bechara Blackwater before he arrived in Barblandia and was able to make peace. Bechara Blackwater and his crew were reasonable pirates who decided that if Barbie could sail in a boat that glorious, then Barblandia must be protected, not attacked. Barbie was left with an unsinkable and remote-controlled boat and Brenton's Buccaneers were left with many valuable lessons about machine engineering design and working in a team.

BACKGROUND AND LITERATURE REVIEW

To begin the process of creating a Barbie boat ready to battle, it was important to complete an extensive analysis of the competition rules, what currently existed in the market for products that fit different aspects of the competition requirements. It was also immensely important to use surrounding resources and background knowledge from other classes to brainstorm ideas for all components of the boat.

As far as the design competition rules go, the most important research that could be done was consulting with the judge. When it came to possible Barbie seats, the best information we found out was that as long as the Barbies could be easily placed by the judges, almost any chair would be considered acceptable. Another key aspect to understand was that come competition day, the rules are left to the judges' interpretation. This translates to building a boat that is solid regardless of the opponent and situation.

The best model of a functioning boat that fell into the size limitations for the competition was in the toy market. For many decades, remote-controlled boats and cars have been enjoyed by people of all ages. Firsthand analysis of a toy's remote-control system was possible due to the pre-competition ownership of a remote-control Jeep from one of the members of Brenton's Buccaneers (see appendix). The Jeep's steering system was controlled using a simple two bar linkage and a servo motor. When the servo would go left, both the front tires would turn left. When the servo would turn right, both the front tires would also turn right. This allowed for a simple understanding of how a servo could be implemented into the boat's steering system to turn the direction the DC motor shaft and propellers were facing.

To understand the buoyancy of the boat, the best reference was MECH 342: Fluid Mechanics for Mechanical Engineering Students. The starting point was knowing that for an object to float, its weight force must be balanced by the upward force of the water it is sitting in. Comprehending this fact is the whole basis behind the hull construction. Whatever weight the boat had must be counter balanced by the total surface area of the boat. The greater the surface area the boat had in contact with the water, the more upward forces there were on the boat to keep it afloat. However, if the surface area was too large, the boat would not sit deep enough in the water to be stable. To account for this, the hull can have a weighted bottom. With these factors kept in mind, a strong foundational understanding of making an unsinkable boat was created.

The next area of research consisted of brushed versus brushless motors. Brushed motors are composed of a rotor, stator, brushes, and a commutator. The brushes maintain physical contact with the commutator to control the direction of current flow in the motor windings [3]. This design simplicity not only results in cost-effectiveness but also lends itself to a degree of water resilience. The brushes, though not completely waterproof, possess tolerance to water exposure. The inherent self-cleaning action of brushes can help displace minor debris or moisture that might enter the motor during operation. This simplicity and moderate water resistance make brushed motors suitable for applications where cost is a crucial factor, and where the operating conditions involve occasional exposure to water, like in basic aquatic hobbyist setups. In contrast, brushless motors are characterized by a more complex design involving electronic controllers (ESCs) and lack physical brushes and a commutator. While they offer advantages in terms of efficiency and precise control, they tend to be more susceptible to water-related issues due to the presence of delicate electronic components. However, manufacturers often produce

waterproof or water-resistant brushless motors for specialized applications, incorporating protective measures such as sealed casings, coated electronics, and sealed bearings. These enhancements aim to mitigate the risks of water damage, making brushless motors suitable for more demanding applications in aquatic environments where precision and efficiency are paramount, such as in high-performance marine models, underwater drones, or professional-grade aquatic vehicles.

In navigating the tumultuous waters of the Barbie Battle Boat Design Competition, the choice of battery technology emerged as a pivotal decision. Each option carried its distinct set of pros and cons [5]. Lithium-ion (Li-ion) batteries showcase formidable attributes, boasting high energy density and a lightweight profile, making them ideal for performance-driven applications. However, the associated costs and temperature sensitivity raise considerations for meticulous project budgeting and environmental conditions. Lithium-polymer (Li-Po) batteries, with their flexible form factor and high energy density, offer versatility in design. However, potential safety hazards and a shorter cycle life warrant careful handling. On the other hand, nickel-metal hydride (NiMH) batteries present a more cost-effective solution with enhanced safety features and a wider operating temperature range, although at the expense of lower energy density and added weight. When assessing waterproofing considerations, Li-ion and Li-Po batteries demanded extra precautions, necessitating protective enclosures to shield against water damage. In contrast, NiMH batteries exhibit a greater tolerance to moisture, providing a relatively more resilient option in damp conditions. Designers in the competition must weigh the trade-offs between energy density, weight, cost, and safety to arrive at an informed decision aligned with the specific demands of their Barbie battle boat. As the boats set sail in the competitive arena, the chosen battery technology not only powered the vessels but also shaped the strategic landscape

inside the hull. As a result, this influences performance, longevity, and ultimately the prospect of claiming victory in the competition.

Arguably the most important area explored in the journey to create the best boat possible was waterproofing. Since the end product would have to be able to remain waterproof for up to three minutes per round, the design of the boat needed to include safe housing for all of the electronics and watertight seals. After researching intensely, an outboard motor system was decided upon. The only thing that needed to be waterproofed was the exit the shaft was coming out of, and the container of the motor itself. An outboard motor also left more room inside the boat for the other electrical components, weapon, and weighting system [2]. Sealants were another important material that needed intense research to guarantee water could not be introduced to the interior of the boat. The best research done on this was going into ACE Hardware and consulting with trained professionals about their recommended sealants. Once a plethora of options was given, the list was narrowed down by carefully examining the back labels of every product and comparing them to one another. Ideally, a sealant would be affordable (less than \$20), applicable on multiple surfaces (including PLA and wood) and would have a fast-curing time (two hours or less). With these criteria set in mind for waterproofing, the boat was set to be waterproof, and the electronics need not fear being exposed to water.

Outside of the direct material and components going onto the boat, more knowledge was required about the engineering design process and division of labor amongst a team. Insight was acquired from attending in class lectures and reading the required texts about properly optimizing time and energy throughout the engineering design process. Overall, the design process is an iterative process that starts with steps to understand the problem at hand, continues with the process of generating ideas, and ends with the implementation, evaluation, and

improvement of the prototype [1]. Idea ranking techniques like Pugh Scoring Matrices and Morphological Charts were very helpful in decision making throughout the design process. While the on-paper guidelines for the engineering design process were clearly provided, the actual understanding about the non-technical skills required to successfully create a boat were learned through trial and error.

Something that cannot always be researched but must always be considered in engineering is the ethical implications of the decisions made along every step of the way. For example, a rule stated no money spent in testing counted towards the final budget of the boat if it was not physically brought to competition that day [4]. While the budget was designed to be affordable to every group, any group with access to more money automatically had more flexibility for mistakes. To combat this potential unfairness, willingness to share resources was a non-negotiable trait every team member possessed throughout the competition. Sharing equipment allowed for no boat to fail due to less access to resources. Sometimes, finding loopholes in rules and regulations may benefit an individual project, but it hurts the bigger picture. Prior internship experience from multiple group members confirms this statement as a fact. In business and in life, it is never ethically right to bend rules if it ends up harming another group unintentionally.

BOM

Item #	Name	Qty	Price	Source
1	1"x12" Pine Common Board	6ft	\$16.5	Home Depot
2	Liquid Nails	1	\$3	Home Depot
3	12V 1000RPM Motors	2	\$3/motor	Alibaba
4	Window/Door Gasket	1 meter	\$0.50/meter	Alibaba
5	Arduino Joysticks	2	\$12 (free due to accessibility)	Mechatronics Lab Amazon
6	9V DC Adapter	2	\$0.10	Alibaba
7	¼" Metal Plate	100	\$41.50 (free due to accessibility)	EMEC McMaster-Carr
8	¼" Steel Rod	20"	\$3/24" (free due to accessibility)	EMEC McMaster-Carr
9	12V ESC	2	\$8/ESC	Alibaba
10	3Ah 20V Lithium-Ion battery	1	\$17/unit	Amazon
11	Galvanized Steel Sheet	2'x2'	\$10	Home Depot
12	Elegoo Arduino	2	\$27.89/Arduino (free due to accessibility)	Dr. Bechara Amazon
13	Solderless Breadboard	2	\$16.00/breadboard (free due to accessibility)	Dr. Bechara Amazon
14	Polylactic Acid (PLA)	3kg	Free	CSU
15	Marine Weld	4	\$8/container (free due to safety precautions)	Home Depot
16	FlexSeal	1	\$17 (free due to safety precautions)	Home Depot

17	Gorilla Glue Silicone	1	\$6 (free due to safety precautions)	Home Depot
18	9V Battery	2	\$1/battery	Alibaba
Total Cost (excluding free/accessbile to all material): \$71.20				

DESIGN PROCESS

With the background information that we had gathered from a variety of different sources detailed above, Brenton's Buccaneers could then look at designing and testing hull designs, propulsion systems, comfort and protection for the passengers, and weaponry. Exploring the methods taught throughout lecture and textbook materials to guide the process of preparing an optimized product.

INITIAL DESIGN CRITERIA (reference to design documentation-)

With the guidelines set in the Competition Description non-negotiable requirements were set early in the design process. The rules for the boat were to be considered customer requirements for our final product which included: maximum boat dimensions, requirement to be a deck boat, ways a barbie could be seated, an International Load Line, safety and testing requirements, restricted materials and weapons, and finally, a budget that must be adhered to.

Following the guidelines and customer requirements presented in the description, a breakdown of these requirements in the scope of the project was necessary to begin designing. In this competition waterproofing and the functionality of the boat was decided to be of utmost importance. A boat that was deemed immobile would be considered eliminated and immediately fall short of the customer requirements. Brenton's Buccaneers knew that this would require multiple tests and debugging of the propulsion system. The boat partaking in 3-minute rounds in a pool, waterproofing from top to bottom to protect the functioning components was given priority to endure as long as possible. We also wanted to have a strong weapon for the battle to demobilize other boats and sustain minimal damage.

After consideration of the project and deliberation among the team our initial design would focus on having a sturdy and watertight hull that would incorporate protection for the

barbies along with easy-to-use propulsion and weapon systems. It was also chosen to focus on addressing the immediate qualifications for elimination in this design project. Brenton's Buccaneers decided that the initial design criteria should include a sturdy boat with a hefty weighting system for satiability.

BRAINSTORMING AND IDEATION (In Hindsight, Retrospective Analysis)

Brenton's Buccaneers began to brainstorm and ideate in the first week of team formation. Initial brainstorming was used to form our initial boat design. As we continued to build and develop the boat, we returned to this process repeatedly to tackle design issues we encountered. We broke our brainstorming into two major categories that could further be divided; weaponization and boat design.

When Brenton's Buccaneers first brainstormed the boat, we set to address an overarching question, "How will other teams build their boats". We unanimously agreed that most boats will have a 3D printed hull. A 3d printed hull would likely be thin, and consist of multiple prints, joined with adhesives such as super glue. Several members of Brenton's Buccaneers are experienced with 3D printed parts that consist of multiple prints and are aware of the drawbacks that come with this approach. Glued 3D prints often break apart if enough force is applied, such as pulling or dropping the components. Thus, our weapon would focus on breaking apart 3D printed boats. With this concept, we began the brainstorming process.

We set a timer for 5 minutes, and every team member jotted down as many weapon ideas as possible. When time was up, we passed our ideas around in a circle, and every team member would select their three favorite ideas that each other team member brainstormed. Many ideas were brought forward. Harpoons, circular saws, projectile launchers, flamethrowers (later

banned), battering ram, and many more ideas were thought of. Alas, one weapon stood out from the rest. The rotating hammer. Thus, Brenton's Buccaneers found their footing.

The brainstorming for our hull revolved around optimizing our hammer. There were two fundamental equations we had to be conscientious of, Newton's third Law 1, and the rotational equivalent of Newton's second law 2. In summary, the force our hammer would exert on another ship would be equally applied to our boat, and our rotating hammer would cause our boat to rotate. As the police chief, Martin Brody said in jaws, "We need a bigger boat" We decided our boat would need to be heavy and durable. Our ideation process for the hull centered on methods we could use to build such a hull. The rotation of the hammer would decrease the more the boat is subsurface. We could either use an incredibly dense material, or have internal weights, the hammer will make the boat front heavy, so we need to weight the back of the boat. We toyed with the idea of a steel boat, but we landed on a thick, wooden hull that ran deep and would be weighed down with either a weighted keel or hull to act as a pendulum to keep the boat from tipping.

HARDWARE AND SOFTWARE INTEGRATION

The integration of components consists of four phases, each representing a material and movement of energy, material, or information. The first three phases are all integration of various types of hardware, beginning with the boat construction. The boat is constructed of wood and sealed with marine body filler. The boat is concerned with the movement of material, as it displaces water from the area it rests within the pool.

The next phase of integration is that of fixtures. These fixtures are 3D printed and serve to build a connection between the ship and the mechanical instruments within. Fixtures take

shape in the form of the barbie seats that keep the barbies protectively encompassed within the ship. The 3D printed motor housings and propellers are also examples of fixtures. The propeller demonstrates the flow of material as it moves water to propel the boat. The motor housing is of the flow of energy, as they attach the motors to the boat, to allow energy to flow. The third phase, the last phase of hardware is that of circuitry. In the circuitry phase of integration, the motors and circuit board move energy in and throughout the boat so it may function as intended. The motors, wires, and battery are key components.

The fourth and final phase is software integration. The software is integrated with the circuitry component and is concerned with the flow of information. The software phase begins with code that is uploaded to an Arduino within a controller. The code is then sent through a transmitter to a receiver that is within the boat. The code then powers the circuitry to activate motors, thus allowing us to control the boat.

ITERATIVE PROTOTYPING AND TESTING SOLUTIONS (Evolution of Design)

A crucial component within engineering is iterating designs based off the evaluation of prototypes throughout the development of a product. Throughout this design competition, multiple prototypes were created to evaluate ideas and concepts. Each prototype and its test highlighted both the shortcomings and successes of the concept or design and much of the final product was constructed in this manner.

Beginning with the hull design (shape, size, and material) it was iterated and constructed through testing and evaluation. As determined by the team plywood was used for the initial iterations of the boat's hull. The first and second version were built to test construction methods and shape. Through testing it was determined that a table saw was the most efficient and precise

way to cut the pieces. Simple geometry was both easy and effective, and wood glue was sufficient to hold the pieces together. As the group suspected, a boat with a wider bottom was found to be more stable than a more pointed bottom when sitting in the water. A third iteration was constructed with these concepts in mind as a suitable design for the competition. However, this boat was constructed to meet the max size detailed in the project description, 14" port to starboard and 20" bow to stern, not allowing for other components that were desired nor included enough width to allow for a reasonable redesign of the internals. A fourth and final iteration was constructed to meet all the requirements that we felt necessary for the hull to achieve.

Another design component that was seamlessly integrated was our method of seating the barbies in the boat and preventing their maiming or forceable removal from the craft. With one of the major winning conditions being Barbies falling overboard or the serious injury of one or both Barbies it was believed to be of the upmost importance in preventing these scenarios. Figure #1 shows a simple cardboard design. It was demonstrated that having a backrest that rose above the barbie's head and a lap bar that only allowed for the barbies to move in a vertical direction was sufficient at keeping the barbies in place.



Figure #1: First Prototype of Barbie Seating

With further models utilizing PLA and 3-D printing, a footrest that prevented horizontal freedom of the legs was implemented. Although the testing that occurred was simple, it was concluded that having a shelf over the barbie's waist and preventing side to side movement of their legs, shown in Figure #2 was effective to protect the Barbies from any foreseeable attacks.

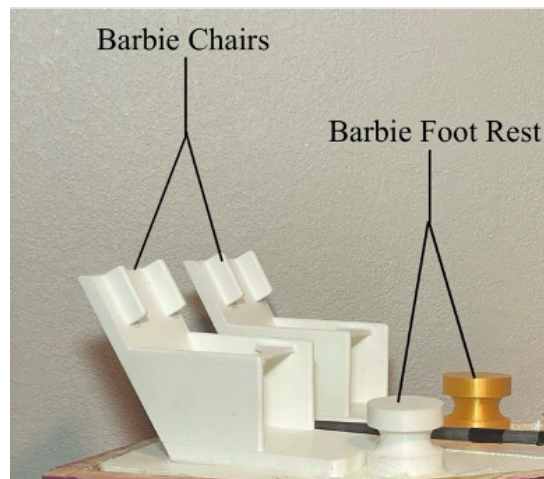


Figure #2: Final Design for Barbie Safety

With movement being grounds for immediate disqualification, many different prototypes were developed in the pursuit of an efficient propulsion system. The initial design was based on an outboard motor system, where a vertical DC motor and shaft with bevel gears would spin the propeller and a servo motor attached to the DC motor controlled the movement. During testing it was proven that the servo motor would rapidly change with input from the controller extremes and waterproofing a mechanism would be more challenging than initially intended. The next prototype was based around mounting two propellers internally and had the same beveled system to attach the propellers on the sides of the ship. Though testing this concept was found to be problematic in the aspect of waterproofing as a spinning shaft would have to go through the hull of the ship. With these prototypes in mind, two stationary outboard DC motors mounted to the base of the boat were determined to be the best solution. This design only required waterproofing of the motors and how they attached to the ship, as holes were needed for programming and control.

In hindsight, we believe that the function of the propulsion system would have exceeded what Brenton's Buccaneers could have accomplished. Waterproof motors could have been utilized in our design allowing for waterproofing of the ship altogether to be more streamlined. A similar mounting system would be used where both motors are attached to the bottom of the boat with wires running into the hull and an on/off mechanism to allow for steering of the ship. More testing of the movement would have also been allocated as the propulsion system functioned, but still had room for improvement in the range of the controller and precision of the movements.

The biggest shortcoming of the design process was our weapons system. The concept of a spinning club that we would use to knock opponents around the pool was decided to be the pursued weapon. The first iteration involved a motor that would spin a metal rod that could be

mounted to the ship and controlled with a button. Upon examination it was determined that there needed to be more force on impact to better effect lighter boats. The second prototype was very similar but involved a much thicker and heavier rod. Testing this in constraints of the hull demonstrated that a longer rod was needed to accomplish our goals. A rod and counterweight system were created to reach the required length while still maintaining balance.

The spinning club weapon was not included in the final design due to uncertainty in its performance and complications that would arise with ensuring a waterproof boat. Insufficient testing on the weapon system was conducted due to complications with 3-D printing the mounting system and logistics attaching the weapon to the boat. Attaching the weapon to the boat would have induced many complications in the waterproofing between the deck and the main body of the ship. Brenton's Buccaneers concluded that for safety reasons and concerns for the function of the rest of the boat, the spinning weapon was not included in the final design.

In retrospect, a few key changes needed to be made regarding the boats' weapon system. Testing the weapon constantly through every prototype would have been informative in both the functionality as well as the changes that needed to be made and potential downsides to the weapon design. When designing the weapon there should have been recognition of how all parts would fit together and how each component of the boat would interact with every other component.

Given the opportunity to redo the project with the knowledge that we have today, Brenton's Buccaneers would have utilized many of the organization and decision-making tactics learned during the semester. The use of a GANNT chart displayed in Figure #3 would have ensured that Brenton's Buccaneers stuck to a schedule that allowed ample time for testing and redesign of the prototypes. Figure #4 shows an estimation of what occurred over the course of

the semester. As shown little to no time was left for testing components within the boat, which shows why the boat fell short of the initial criteria.

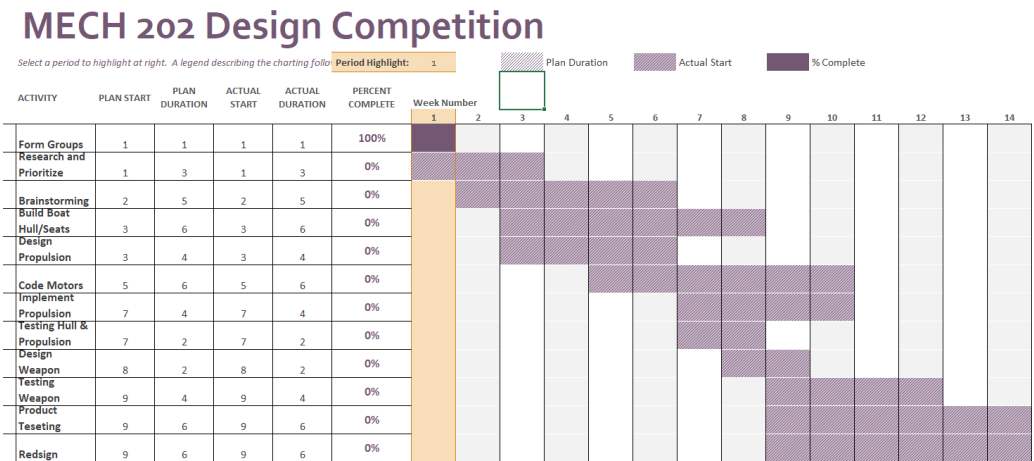


Figure #3: Ideal GANNT Chart

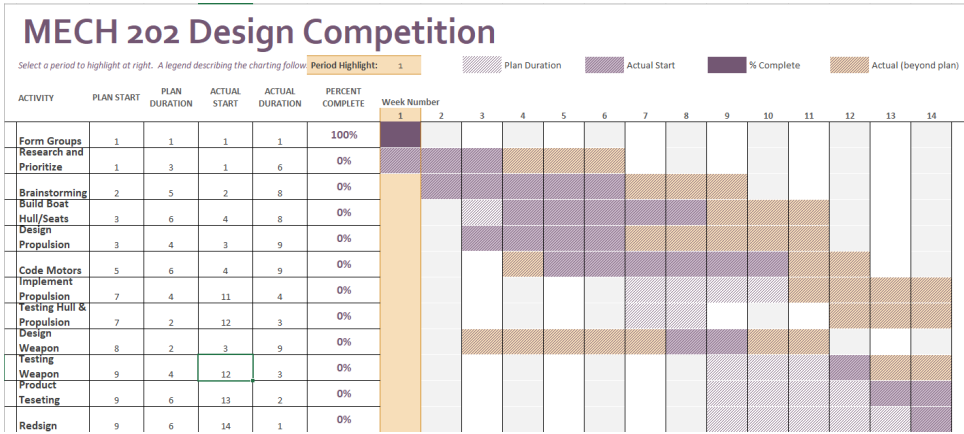


Figure #4: Estimation of Actual GANNT Chart

Throughout this design process Brenton's Buccaneers learned that when designing a new product just how important testing is. Many of the areas that fell short of our design criteria were due to a lack of testing and neglecting the importance of brainstorming solutions throughout the

duration of the project. It was discovered that ideas, concepts, and prototypes failing are the path to having a successful final design.

FINAL SOLUTION

The final solution, shown in Figure #6, is the product of every team member's brainstorming, prototyping, testing and integration of components. The hull was the fourth iteration of boat designs. It follows the same geometry of the second and third hull iterations, but with adjustments. The ship is shorter and taller, which we concluded was essential for our design to work.



Figure #5: Final Boat used During the Competition

An ongoing problem for our integration of components was combining fixing the rotating hammer to the deck while also keeping it waterproof. To change batteries in between rounds, our deck is removable. This also lets us adjust the circuitry that is shown in Figure #7.

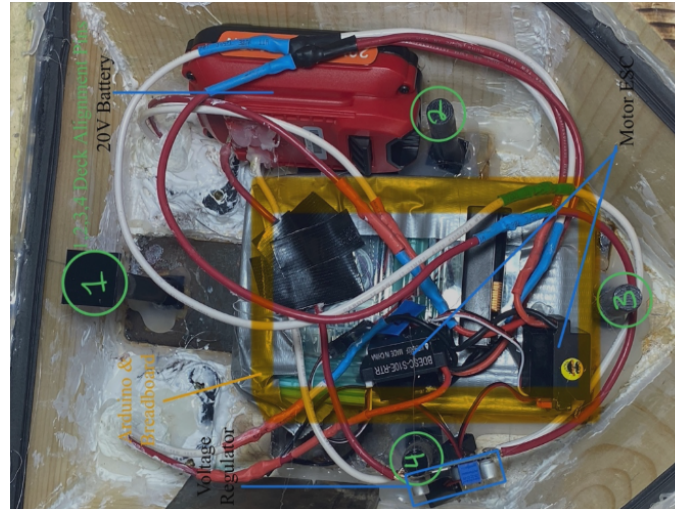


Figure #6: Circuit's Within the Boats Hull

The weapon we implemented for competition was a battering ram. This worked well with our removable deck design, and the final hull. With a shorter hull, we can extend the battering ram a great distance over the front of the ship to directly attack barbies. The added height of the ship also allows us to attack over most other ships.

In the final solution, we opted for a weighted hull over a weighted keel. We used multiple steel plates that we could cover throughout the bottom of the boat. These weighted plates prevented our ship from capsizing as they essentially turned our boat into a pendulum. Using steel plates for our hull, we could disperse the weight across the bottom, but also adjust them as needed to counteract heavier sections of the boat. The heavier boat also improved our boat's attack, as the heavier ship we have, the more force is exerted when we collide with another boat.

Another challenge we encountered is friction between gears for our motors. We wanted to keep our motors safely sealed within the boat, which is why a larger hull was optimal. However, the motors had difficulty rotating the 3D printed gears. Thus, we built motor housings that are fixed outside the hull. Through this motor housing, we cut down the energy loss in our motors and had a more powerful boat.

Altogether, our ship was reliable, durable and effective. On December 2nd, the results of our semester long efforts were showcased as we successfully defeated our first opponent. In the second round, we sadly lost the tiebreaker vote by a close margin.

ENTRPRENEURSHIP AND MARKET ANALYSIS (optional- posters for style comp, competitors)

Our product prototype was an excellent model. However, if we were to sell it, we would need to update our manufacturing processes for mass production. As it stands, recreating our ship requires around 5 hours of carpentry, 2 hours of body work, and 5 hours of assembly, this does not include any time for glue drying or 3D printing. To mass produce our product, we need better machinery.

Firstly, rather than cutting everything with a miter saw, we would utilize a CNC mill to cut all our wood with G-code. The Motor housing and other 3D printed models would be upgraded to carbon printing. Carbon printing is faster, more precise, stronger than PLA, and overall, creates a better product. The main manufacturing process that would remain up to manual labor, is the installation of circuitry. All in all, our weaponized boat could be produced for the masses.

To be successful entrepreneurs, we must hone our sales skills. We believe Shark Tank would be a great platform to launch our product as a rival product to Barbie. The total value of the global toy industry is 128 billion USD, (<https://www.fortunebusinessinsights.com/toys-market-104699>). Mattel, the manufacturer of Barbie, made up 42.3% of the total share of 2023 Q3, (<https://csimarket.com/stocks/competitionSEG2.php?code=MAT>). As Barbie is a large, profitable industry, our approach to enter the toy industry is to sell our product idea to Hasbro, or another competitor of Mattel. Our product marketing would directly compete against Barbie, and openly demonstrate that we make a more exciting product than the Barbie line of ship.

ETHICAL, COMPLIANCE, AND SUSTAINIBILITY (Real boat and what it would take, Toy and what it would take, environmental impact/ waste)

Ethical considerations, regulatory adherence, and sustainability measures cannot be ignored and must be components guiding the trajectory of the competition. In the context of RC boat design, the ethical compass extends beyond conventional safety norms to encompass the well-being and engagement of the Barbie boat captains. The waterproofing of electronics emerges as a paramount ethical consideration, recognizing that wet electronics pose potential hazards to users sharing the pool. The boat competition mandates a thorough approach to waterproofing, ensuring the safety of both participants and spectators. Beyond this, ethical testing practices involve a commitment to fair competition, emphasizing the boats' resilience to battle rigors without compromising the safety and integrity of the Barbie boat captains. Brenton's Buccaneers even faced an ethical dilemma when there were 24 hours left and no weapon system was mounted and tested in the finalized hull. **Instead of choosing to risk everyone's safety and including an unregulated and untested weapon, it was decided to omit a weapon from the boat.** **One group even chose not to compete due safety concerns about exposed wires from their boat.** This ethical framework fostered an environment where creativity met responsibility, and thrilling competition aligned harmoniously with user safety.

Regulatory considerations define the structural boundaries of the RC boat competition, with size dimensions serving as principal constraints from the project's inception. Participants navigated these dimensions with precision, ensuring that their designs adhered to the established criteria. Moreover, the strategic placement of Barbie boat captains becomes a regulated element, emphasizing not only comfort but also the visibility of the Barbies' shoulders at all times. Another critical regulation involved an accurate international load line, functioning as a gauge

for boat buoyancy and preventing unsafe sinking scenarios. This multifaceted regulatory framework guided designers in crafting boats that are not only innovative and competitive but also compliant with established rules. As a result, a level playing field was created for a thrilling battle boat arena.

The RC boat competition incorporated sustainability with the entertaining dynamics of the battle boat arena. Designers are encouraged to scrutinize the environmental impact of their boats, considering factors such as the use of recyclable materials and minimizing plastic consumption. Sustainable practices became inherent to the competition, aligning innovative design with ecological responsibility. As participants explored the balance between thrilling competition and environmental stewardship, the competition became a showcase not only of engineering ingenuity, but also of a commitment to a sustainable future.

The Barbie Battle Boat Competition's ethical, regulatory, and sustainability dimensions converged to shape an event where creativity is tempered by responsibility. Participants, as stewards of both competition and environment, navigated these multifaceted considerations, ensuring that the RC boat battle transcends mere entertainment to become a model of responsible and sustainable engineering. This is a mindset that will carry on in each engineer's mind in future endeavors they choose to take.

TEAM DYNAMICS AND REFLECTION (learned something, would have done different)

To represent a real-world design team, the students were unable to choose their entire group and paired with non-acquainted students to work together. In the workforce, team dynamics, communication, and conflict resolution are vital for success. For Brenton's Buccaneers this would be one of the greatest learning experiences of the semester. As a team we were able to reliably have full attendance during our agreed upon meeting times. This proved very important for getting to know our other members and building upon the team dynamic in a productive way. Our biggest struggles throughout the semester were communication of individual expectations and ideas. Throughout the semester Brenton's Buccaneers had to learn to collaborate with new people with different learning styles. We also had to learn to accept different personalities with different communication preferences. This was challenging as team members' expectations within the group were often misconstrued and conflict resolution became an important learning experience for everyone involved. The battle boat project required teamwork throughout the course of the semester. Unfortunately, much of the work was divided among individuals within the team which was unproductive to the team dynamic. Many decisions about the project were made without mutual understanding or agreement of the team as a whole. The lack of communication caused many of these changes to be received in a poor manner that negatively affected the team morale as a whole. Unclear expectations within the group caused a couple of shortcomings among the members individual's focuses on the boat. Meshing individual designs proved especially challenging in this project as seamless integration would have required better communication and design sharing among the team. This project has provided insight into what is required to work as a team towards a certain goal.

In retrospect, the team dynamic should have been a larger focus within Brenton's Buccaneers. The initial step in this design process should have been building a team image, working to lay out clear methods, and expectations for communication and idea generation. We believe that this would have allowed free flowing ideas and a concise team effort. Brenton's Buccaneers learned many key concepts in team development and maintain a positive team dynamic that will certainly be useful in future design groups.

CONCLUSION

When challenges arose throughout the semester, the main motivator that kept this group going was to create a boat this team could be proud of in competition. Given all the challenges necessary to overcome to get to competition day, it is safe to say a beautiful and practical boat was in the pool on December 2nd. Brenton's Buccaneers never lost a Barbie overboard, were always able to control the boat, and never sunk. The goal to be an indestructible, maneuverable vessel was inarguably accomplished.

This design competition speaks to the broader challenges in engineering because just like this class, things do not always go as planned. In industry, group dynamics and meeting deadlines will always be a challenge. The important thing to remember is to never lose sight of the goals behind projects and try to find common ground amongst everyone in the group. At the end of the day, hard work and perseverance will always shine through, so it is important to never give up and keep rowing forward.

REFERENCES

[1] A. Saterbak and M. Wettergreen, *Introduction to Engineering Design*. Springer Nature, 2022.

[2] “Boat Engine Types Explained,” *BOATERexam.com*®.

<https://www.boaterexam.com/boating-resources/boat-engine-types-sizes/>

[3] PE, “What Is the Difference Between a Brushless and a Brushed Motor?,” *Power Electric*.

<https://www.powerelectric.com/motor-resources/motors101/what-is-the-difference-between-a-brushless-and-a-brushed-motor>

[4] S. Bechara, *Doll’d Up Nautical Knockdown: Where Fashion Meets Splashin*. 2023.

[5] Webstaurant, “Types of Batteries,” *WebstaurantStore*.

<https://www.webstaurantstore.com/guide/923/batteries-buying-guide.html>

APPENDICES

```

37
38   data.Lval = analogRead(Lpot); // Scale the potentiometer value to a range of 0-255
39   Serial.println(data.Lval);
40   data.Lval = map(data.Lval, 0, 1023, 0, 50);
41   data.Lval = map(data.Lval, 0, 50, 550, 2650);
42
43
44
45   // Send the data structure to the receiver
46   radio.write(&data, sizeof(Data_Package));
47
48   Serial.println(data.f);
49   Serial.println(data.Rval);
50   Serial.println(data.Lval);
51
52   delay(10);
53 }
54

```

```

Transmitter.ino
1  // Include Libraries
2  #include <SPI.h>
3  #include <RF24L01.h>
4  #include <RF24.h>
5
6  // Create an RF24 object
7  RF24 radio(7, 8); // CE, CSN
8
9  const int Lpot = 0; // Define the analog pin used to connect the potentiometer (Left)
10 const int Rpot = 1; // Define the analog pin used to connect the potentiometer (Right)
11 struct Data_Package {
12     int Rval; // variable to read the value from the analog pin (Right)
13     int Lval; //Left Linear potentiometer Variable
14     char f[36] = "Brenton is still figuring life out";
15 };
16
17 Data_Package data; // Create an instance of the Data_Package struct
18
19 // Address through which two modules communicate.
20 const byte address[6] = "00001"; // COULD use an address like this...
21
22 void setup() {
23     Serial.begin(9600);
24     radio.begin();
25     radio.setChannel(2471); // Set channel number here, between 0 to 125
26     radio.openWritingPipe(address); //code from the testing code
27     radio.setPALevel(RF24_PA_HIGH); //sets the power level of transmitting/receiving
28     radio.stopListening();
29 }
30
31 void loop() {
32     // Read the value of the potentiometer
33     data.Rval = analogRead(Rpot);
34
35     data.Rval = map(data.Rval, 0,1023,0,50);
36     data.Rval = map(data.Rval, 0, 50, 550, 2650);

```

```

Reciever.ino
1  // Include Libraries
2  #include <SPI.h>
3  #include <RF24L01.h>
4  #include <RF24.h>
5  #include <Servo.h>
6
7  // Create an RF24 object
8  RF24 radio(7, 8); // CE, CSN
9
10 Servo Rservo; // Create servo object to control a servo
11
12 Servo Lservo;
13
14 //const int motorPin = 5; // Define the PWM pin connected to the motor
15
16 struct Data_Package {
17     int Rval=0; // Variable to read the value from the analog pin
18     int Lval = 0; //variable to read the value from the linear potentiometer pin
19     char f[36]="";
20 };
21
22 Data_Package data; // Declare an instance of the Data_Package struct
23
24 // Address through which two modules communicate.
25 const byte address[6] = "00001"; // COULD use an address like this...
26
27 void setup() {
28     Serial.begin(9600);

```

```

29
30 radio.begin();
31 radio.setChannel(2471); // Set channel number here, between 0 to 125
32 radio.openReadingPipe(0,address);// this is from the testing code uncomment below code once you delete this
33 //radio.openWritingPipe(0xD0A7A7A7A7LL);
34 radio.setPALevel(RF24_PA_HIGH); //sets the power level of transmitting/receiving
35 Rservo.attach(3,1000,2000); // Attaches the servo on pin 9 to the servo object
36 Lservo.attach(5,1000,2000); // Attaches the servo on pin 9 to the servo object
37
38 // Set module as receiver
39 radio.startListening();
40 }
41
42 void loop() {
43     // Check whether there is data to be received
44     if (radio.available()) {
45         radio.read(&data, sizeof(Data_Package)); // Read the whole data and store it into the 'data' structure
46
47         //Serial.print("val"); // This should present the value of the potentiometer after it's already been adjusted for the servo
48         Serial.println(data.Rval); // Print the value read from the 'data' structure
49         Serial.println(data.Lval); //Print the Value of the Lpot
50         //Serial.print(data.f); // Should present "Test" in the serial window
51         //Serial.println(data.f); // Print the value of 'f' from the 'data' structure
52         Rservo.write(data.Rval); // Sets the servo position according to the scaled value (Move to receiver)
53         Lservo.write(data.Lval); // Sets the servo position according to the scaled value (Move to receiver)
54     }
55     delay(10);
56 }
57

```

[#] Newton's third law $F_1 = -F_2$, $m_1 a_1 = -m_2 a_2$

[#] Rotational equivalent of Newton's second law $\text{torque} = mr^2 a$